

you can tell the software to either prompt you for instructions or just continue or cancel execution of the task.

STEP 4 Define triggers: Triggers are necessary in Macro Angel to set off an event that has not been scheduled to run at a particular time. There are several types of triggers for you to choose from according to the task you want to set off. **Hotkeys:** Setting up hotkeys makes sense when you want to launch a task manually. An example of a good situation to use this feature is while launching an application or a Web site.

Magic keys: Using this trigger, Macro Angel can launch a task when it detects that a certain string of text has been typed into any application on your system.

Appearance of a window: This allows you to execute your task on the appearance of a specific application window. You can specify this window from the list that appears when the checkbox activating this kind of trigger is selected.

Although one trigger is sufficient, in most cases you can assign multiple triggers to a single task if you want to launch it in a range of circumstances.

STEP 5 Task Steps: This is the most important stage because 'Task Steps' are the key to helping Macro Angel provide the functionality to simulate your actions on your PC. Each 'step' involves the use of one of many types of actions that are possible on a PC. Once a Macro Angel task is triggered, the program moves through the steps defined during the task creation.

This is the part that requires the most planning. Not only is it necessary to choose the right task steps, but you need to ensure that these are arranged in the correct sequence so that everything goes off as planned.

Let the action begin

One of the most critical steps towards achieving a properly functioning macro is to understand the actions recognised by the software and using them appropriately. While Macro Angel and other similar software support a large number of actions, you are likely to use some of them more than the others. It is essential to get a general understanding of these action types and what they do before you actually begin creating a macro.

Sending keystrokes: This is the first button in the 'Task Steps' pane and clicking

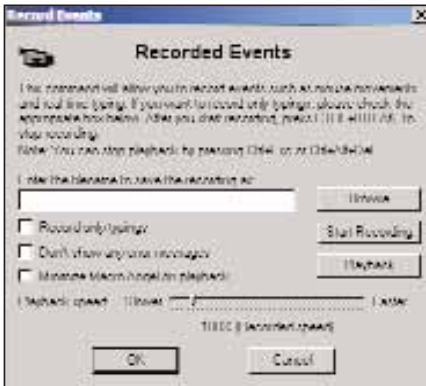


A sequence of action items are executed to perform a task

on it will take you to the 'Send Keystrokes' menu. It lets you enter text that you need to send to a particular application. The application window can also be specified so that the text isn't sent to the wrong location. For this to work properly, the application window needs to be 'in focus' or activated.

Miscellaneous actions: This collection of actions lets you perform tasks such as starting an application and opening a document or Web page. The 'Show a Message' action displays a Windows-style dialog box with the text you specify, which is meant for providing feedback during a task's execution cycle.

The most versatile and useful tool however is the 'Record Events' option. This allows you to record the movements of your mouse as well as keystrokes to a file, which can then be played back through a task to simulate the same mouse movements. For example, you can record your mouse movements as you double-click an icon to open a folder. Then just play these mouse movements back through Macro Angel, and you will see the pointer move about on the screen as if you were actually moving the mouse yourself. The program will continue to record events and



Record events and play them back to perform tasks

save them to a selected file until you press [Ctrl] + [Esc] on your keyboard.

Handling windows: Working with different application windows is necessary from time to time, especially when you decide to use the appearance of a window as a trigger for an event. This menu lets you focus on a particular window and then maximise, minimise, resize, close or even hide it.

Flow control: When designing a complex task made of different options, it becomes necessary for you to have some control over the progress of the task from one action step to another. Macro Angel provides us with several flow control tools that make this possible. A Windows-like dialog box can be made to appear, which pauses the task and prompts you to click either a 'Yes' or 'No' button. Depending on your choice the task either proceeds or can be terminated.

Macro Angel also allows the creation of loops that consist of one or more steps. These loops let you repeat a series of actions as many times as you like without your having to create them repeatedly.

However, one of the most powerful features of automation software is the ability to create conditional actions. They are usually used to check whether a previous action has been successfully completed. These conditional 'If/Else/End if' options help to completely automate tasks and completely eliminate the need for user intervention, which becomes necessary if even one step fails to complete correctly.

System procedures: This set of actions gives you control of Windows system level operations such as file and folder management, and program termination. It even lets you reboot your computer when necessary. You can even automate actions such as Backup, which lets you create automatic backup processes that can even compress multiple files into a single ZIP file in order to save space. Other useful file handling actions include 'Copy', 'Move', 'Delete', 'Rename' and 'Findfiles'.

Task management: Often a task might consist of several subtasks that handle dif-



You can even control Windows' system level operations